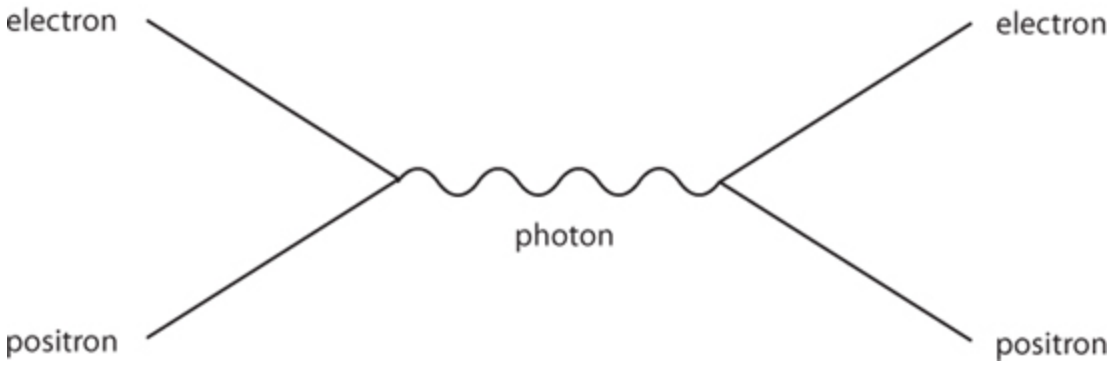


The story we tell based on the Feynman diagrams is just that, a story. It's not literally true that an electron and a positron change into a photon and then change back. For one thing, real photons move at the speed of light, while electron/positron pairs (either the individual particles or the center of mass of a pair of them) do not.

What actually happens is that both the electron field and the positron field are constantly interacting with the electromagnetic field; oscillations in any electrically charged field, such as the electron or positron, are necessarily accompanied by subtle oscillations in the electromagnetic field as well. When the oscillations in two such fields (which we interpret as the electron and positron) come close to each other or overlap, all of the fields push and pull on one another, causing our original particles to scatter off in some direction. Feynman's insight is that we can calculate what's going on in the field theory by pretending that there are a bunch of particles flying around in certain ways.

This represents an enormous computational convenience; working particle physicists use Feynman diagrams all the time, and occasionally dream about them while sleeping. But there are certain conceptual compromises that need to be made along the way. The particles confined to the interior of the Feynman diagrams, which don't either come in from the left or exit to the right, don't obey the usual rules for ordinary particles. They don't, for example, have the same energy or mass that a regular particle has. They obey their own set of rules, just not the usual ones.

That shouldn't be surprising, as the "particles" inside Feynman diagrams are not particles at all; they're a convenient mathematical fairy tale. To remind ourselves of that, we label them "virtual" particles. Virtual particles are just a way to calculate the behavior of quantum fields, by pretending that ordinary particles are changing into weird particles with